Metadata and Image Features Co-Aware Semi-Supervised Vertical Federated Learning With Attention Mechanism

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***Abstract*—Recently, vertical federated learning (VFL) has gained more attention due to its ability to combine various valuable features for better model performance without violating privacy. However, the development of VFL still faces three challenges: 1) insufficient aligned samples, 2) monotone data modality, and 3) simple concatenation-based collaborative training strategy. In this article, we construct a metadata and image features co-aware VFL model for smart healthcare, from which users can obtain convenient and timely online auxiliary diagnostic services. Furthermore, we propose a metadata and image features co-aware semi-supervised VFL scheme with attention mechanism, which aims to jointly train the VFL models by utilizing both metadata and image data from the entire sample space. Specifically, a generative-adversarial method is designed to train the generator to estimate the missing features, so as to expand the training set and improve the accuracy of model. Moreover, using semi-supervised clustering to forecast pseudo labels for the missing part of the active party, we further boost the performance of the VFL model by increasing the proportion of labeled samples. Meanwhile, inspired by the attention mechanism, we establish an entirely novel VFL training mode that emphasizes the underlying effect of metadata features on image features, which makes metadata features highlight crucial areas in the image features to improve the training effect. Finally, the experimental results confirm that our proposed scheme achieves higher accuracy compared with other related works in the VFL scenario.**

***Index Terms*—Vertical federated learning, semi-supervised learning, attention mechanism, bimodality fusion.**

I. Introduction

The digital age in which we currently live is wealthy with numerous kinds of data, which is generated by healthcare sensing devices, smart vehicles, industrial robots, and more [1]. In the meantime, this data is increasing day by day. As a result, recent years have seen a rapid development of data-driven machine learning (ML) technology, which can timely and intelligently extract useful knowledge from data to serve as the foundation for practical applications. ML has been extensively employed in smart clinical decision-making [2], smart assisted driving [3], intelligent manufacturing [4], and other real application fields. However, traditional ML typically involves centralizing scattered user data to one source, which conflicts with the growing awareness of data security and user privacy. In response, relevant laws exist to strengthen data security and privacy protection, such as General Data Protection Regulation. Moreover, this centralized training approach requires transmitting a large amount of raw data which brings high communication overhead.

In this regard, a distributed technique named federated learning (FL) was put forth by Google in 2017, which aims at collaboratively training a global ML model while protecting privacy [5]. Yang et al. classified FL into three categories in literature [6]: horizontal FL (HFL), vertical FL (VFL), and federated transfer learning (FTL). On the one hand, HFL, also known as sample-based FL, generally involves a sizable number of parties employing various samples while sharing the same feature space. To train a global diagnosis model, for instance, various medical facilities in the smart healthcare scenario make use of image features of different patients. On the other hand, VFL is also called feature-based FL, which typically includes two or three parties with the same sample set but various feature spaces. As an example, different medical facilities collaboratively train a diagnostic model with image features and metadata features of the same patients, respectively. In addition, FTL is suitable for scenarios where datasets differ both in sample set and feature space.

In recent years, VFL has gained more attention in academia and industry due to its ability to combine various valuable features for better model performance without violating privacy. VFL is increasingly being applied in practical scenarios such as recommendation systems, finance, and healthcare. For example, literatures [7], [8] and [9] proposed the VFL recommendation system framework to improve the accuracy of recommendations. WeBank builds the risk control VFL model in conjunction with the credit data of customers and the invoice information of partner companies [10]. Besides, some works [11], [12] and [13] applied VFL to healthcare recommendations and clinical

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diagnosis, which improve the efficiency of medical service and alleviate the pressure of healthcare treatment. However, the development of VFL still faces the following three challenges:

1) *Insufficient aligned samples:* Sharing a sufficient number of aligned samples among all parties is a crucial prerequisite for VFL to ensure efficient model performance. In practice, the number of aligned samples among various parties is typically constrained and gets progressively smaller as the number of parties increases. In light of this, if we only focus on the aligned data to train VFL, we would squander a majority of non-aligned samples and be unable to further increase the competitiveness of the VFL models.
2) *Monotone data modality:* So far, most VFL works have mainly studied single data modality such as image, tabular, or text [14], [15], while researches on VFL holding bimodality or multi-modality data are relatively limited, which is of great significance in practical application and is beneficial to enhancing model training effect.
3) *Simple concatenation-based collaborative training strategy:* In the training process of VFL, the representations extracted by all parties are frequently transferred to the active party with labels for collaborative training. Existing VFL techniques, however, almost typically use a straightforward concatenation to combine various representations before feeding them into a neural network for training. This type of direct concatenation ignores the underlying influences between representations. How to design new representation fusion techniques to enhance model performance has become a valuable research area.

Inspired by these challenges, we propose a metadata and image features co-aware semi-supervised vertical federated learning (MIFC-SVFL) scheme with attention mechanism. The main contributions of this article are summarized as follows:

1) We focus on the VFL of two modality features, tabular and image, which has important practical value for disease diagnosis based on both patient metadata and medical images in smart healthcare. Therefore, we construct an intelligent diagnosis model for smart healthcare, which can offer nearby users convenient and timely online auxiliary diagnosis services.
2) We present a metadata and image features co-aware semi-supervised VFL scheme, which aims to jointly train the VFL models by utilizing features from the entire sample space. In order to make full use of the non-aligned samples, we design a generative-adversarial method to train the generator to estimate the missing features by learning the class distribution of the available features, so as to expand the training sample set and improve the accuracy of model training by means of data augmentation. Moreover, since both labels and features of the missing part of the active party are absent, we design semi-supervised clustering to forecast pseudo labels for the missing part of the active party, which further boosts the performance of the VFL model by increasing the proportion of labeled samples.
3) Furthermore, inspired by the attention mechanism, we establish an entirely novel VFL training mode that emphasizes the underlying effect between different modality features. To the best of our knowledge, this study is the first to break the simple representation concatenation in the typical VFL for collaborative training. That is, the federated model is guided by metadata information to concentrate on the particular area of the representation channel, so that it may fully mine the essential information of the image features and improve model training.
4) Finally, the experimental results confirm that our proposed scheme achieves higher accuracy compared with other related works in the VFL scenario.

The remainder of this article is arranged as follows. Section II describes the related works. The network model is constructed in Section III. In section IV, the designed MIFC-SVFL scheme with attention mechanism is illustrated. Subsequently, the experiment results and evaluations are presented in Section V. Finally, conclusions are drawn in Section VI.

II. Related Work

VFL also referred as feature-based FL, utilizes various features distributed among different parties to jointly train federated models for improved performance while preserving privacy. The advantages of this training strategy motivate many researchers to study efficient VFL schemes, which are mainly divided into the following two categories.

The first category, works [16], [17], [18], [19], [20] mainly employed data sharing the same sample ID among different parties, which is also known as aligned samples, to train VFL models. For instance, Liu et al. [16] proposed a VFL framework based on block coordinate descent optimization, which enabled each party to perform multiple local updates before global communication, effectively reducing the communication overhead. In addition, literature [17] constructed a one-shot VFL scheme based on unsupervised learning, where parties can extract representations from raw features through unsupervised learning, and sent them to the active party to train the federated model, leading to a significant improvement in the model accuracy and generalization ability. Similarly, work [18] used autoencoders to extract representations from each party for model training, which achieves corresponding effects in practical applications. During training, however, these VFL schemes typically input the data sharing the same sample ID among different parties, and ignore the non-aligned samples. As a result, these samples that possess potential values are not fully exploited, resulting in suboptimal model performance.

To address the aforementioned problem, the second category, works [21], [22], [23], [24], [25], [26], [27], attempted to exploit non-aligned samples for training in an effort to enhance model performance. Kang et al. [21] estimated the missing features and labels in the non-aligned sample space through cross-view training based on aligned samples, allowing them to train the federated model with data from the whole sample space. Similar to this, literature [22] also enhanced model training by expanding the aligned dataset. Specifically, a common feature space is built using aligned samples, onto which non-aligned samples are projected. Next, the similarity between each pair of non-aligned samples in this space is measured by calculating their distance. Afterward, these non-aligned samples are matched and added to the aligned samples. Finally, the expanded aligned samples are used for training to create a prediction model. In addition to the above works of expanding the aligned dataset, literatures [23] and [24] utilized non-aligned samples to train the local model through self-supervision. As an illustration, authors in [23] employed the knowledge gained by each party in the aligned sample space to jointly train the local cross model, which is then used to direct the local aligned and non-aligned samples to self-supervise the training of local models. Besides, Li et al. [25] successfully leveraged aligned and non-aligned data to train the model through feature imitation and ranking consistency restriction. Although the aforementioned works utilize the non-aligned sample for training to some extent, they still face the challenges that using a single modality of data is insufficient to train the model, and using a simple concatenation-based collaborative training strategy cannot fully mine the hidden relations among features to enhance model performance.

How to fully exploit the hidden relations among different features to enhance training? The attention mechanism may provide some insights. The attention mechanism is a distinctive structure embedded into machine learning models to automatically determine and quantify how much each component of the input data contributes to the output. It has been widely applied in computer vision including image classification [28] and object detection [29] to improve the performance and generalization of models. Its major objective is to imitate how the human visual system focuses on salient parts in complex scenes by dynamically adjusting the weights based on the input image features [30]. Existing attention methods can be divided into three categories: channel attention, spatial attention, and channel & spatial attention. The idea of channel attention was first put forth by Hu et al. [31] who adaptively changes the weight of each channel to identify the primary object of attention. Selective kernel networks (SKNets) [32] applied transformations with various convolution kernel sizes to create the feature maps obtained by various receptive fields, and after that, channel-wise weighting was used to fuse information from all branches. Works [33], [34], and [35] were typical spatial attention mechanisms that adaptively select important spatial areas. Channel & spatial attention adaptively chooses essential objects and regions by taking into account the significance of features in both the spatial and channel dimensions [36], [37], [38].

Furthermore, the above VFL works were based on the premise that alignment between parties has been implemented. Some studies [39], [40], [41], [42] focus on private entity alignment, and these methods can be combined with our scheme, which mainly concentrates on the aspect of model training, to achieve private entity alignment.

III. Network Model

VFL, which employs various features from several parties to train the model jointly while maintaining privacy, is ideally suited for merging patient metadata features and image features to train the diagnostic model collaboratively in the context of smart healthcare. Inspired by this, we construct a metadata and image features co-aware VFL model for smart healthcare. The model is divided into three layers, as illustrated in Fig. 1: the user layer, the edge layer, and the cloud layer. The specific functions and relations among these three layers are described as follows.

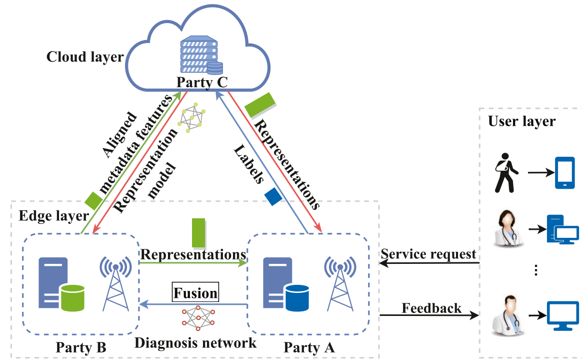


Fig. 1. Metadata and image features co-aware vertical federated learning model for smart healthcare.

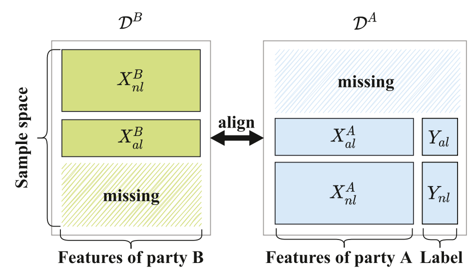


Fig. 2. View of the data distribution of two parties.

User layer: In this layer, both medical professionals and patients have access to the edge layer to request diagnostic services from parties. In order to acquire auxiliary diagnosis results to improve diagnostic accuracy, medical professionals utilize computers to send medical images (such as dermoscopic images and magnetic resonance images, etc.) and patient information (such as age, gender, and cardinal symptoms, etc.) to the edge layer. In a similar manner, patients upload the disease images taken with their mobile phones and fill in the necessary information to the edge layer to get the preliminary diagnosis results and assist them in deciding whether to seek further medical treatment.

Edge layer: Consistent with the typical two-party VFL, the edge layer consists of an active party (party A) with image features and labels, such as a medical institution with abundant experts, and a passive party (party B) with metadata features, such as a health self-check software with sizeable users. Only a small portion of samples of party A and party B are aligned, as seen in Fig. 2. Therefore, in order to fully train the diagnosis network using the data from the entire sample space, we rely on the trusted party C in the cloud layer to complete the missing data of party A and party B. More specifically, the edge layer serves two functions: training the diagnosis network and providing diagnosis services.

1) Training the diagnosis network: For the purpose of estimating the missing features, party A and party B, respectively, upload the labels and aligned raw metadata features to party C. Subsequently, party B receives the representation model trained with the estimated and aligned metadata features on the cloud to extract the representations of the non-aligned metadata samples and then transmits them to party A. At the same time, party A downloads the remaining metadata representations from the cloud, and utilizes representation clustering to estimate the labels of party A's missing samples, it then uses the estimated labels and its pre-trained conditional generative adversarial networks (CGAN) to estimate the missing image features. After fusing the image representations extracted from the local image features with the metadata representations based on the attention mechanism, party A trains the diagnosis network. Finally, the diagnosis network and fusion model are sent to party B to ensure that users can obtain diagnosis results from each party in the edge layer.
2) Providing diagnosis services: When the party in the edge layer receives a request from the user layer, it combines the metadata feature and image feature given by the user, and then returns the diagnosis result to the user. Meanwhile, the samples will be stored in the local database to expand the training dataset.

Cloud layer: This layer contains a party C that possesses powerful computing capability and is trusted by the edge layer. Party C receives the labels and aligned metadata features from the edge layer in order to estimate the missing features of party B using the conditional generative adversarial imputation network (CGAIN), and then utilizes the estimated and aligned features to train the representation model of party B in a supervised manner. After that, the representation network is returned to party B to extract the representations of the non-aligned features, while the remaining representations of party B in party C are returned to party A to predict the missing image feature labels.

The main notations in this article are summarized in Table I.

TABLE I
TABLE OF MAIN NOTATIONS

Notation	Description
$\mathcal{D}^A$, $\mathcal{D}^B$	The dataset owned by party A, B.
$\mathcal{D}_{al}$	The aligned sample set.
D_{al}^A, D_{al}^B	The part of aligned sample set owned by party A, B.
D_{nl}^A, D_{nl}^B	The non-aligned sample set owned by party A, B.
x_i^A, x_i^B	The feature of the i^{th} sample owned by party A, B.
$x_i^{A,al}$, $x_i^{B,al}$	The aligned feature owned by party A, B.
$x_i^{A,nl}$, $x_i^{B,nl}$	The non-aligned feature owned by party A, B.
y_i	The label of the i^{th} sample.
y_i^{al}, y_i^{nl}	The label of the aligned and non-aligned sample.
N^A, N^B	The samples number of $\mathcal{D}^A$, $\mathcal{D}^B$.
N^{al}	The number of aligned samples.
N_A^{nl}, N_B^{nl}	The number of non-aligned samples owned by party A, B.
$\mathbf{M}$	The random mask vector.
$\mathbf{Z}$	The random noise.
$\mathbf{H}$	The hint vector.
$\tilde{\mathbf{X}}$	The feature vector after random mask.
$\bar{\mathbf{X}}$	The feature vector estimated by generator G.
$\hat{\mathbf{X}}$	The final imputed vector.
R^A, R^B	The representation of features in party A, B.
F_A, F_B, F_F	The neural network to extract image, metadata and federated representations.
d_{ij}	The distance between representations of different classes.
c_k	The center point of the k^{th} cluster.
$\tilde{\mathcal{F}}$, $\hat{\mathcal{F}}$	The transformations of two different kernel sizes.
$\tilde{R}$, $\hat{R}$	The representations converted by $\tilde{\mathcal{F}}$, $\hat{\mathcal{F}}$.
u, v	The soft attention vector for $\tilde{R}$, $\hat{R}$.

IV. METADATA AND IMAGE FEATURES CO-AWARE SEMI-SUPERVISED VERTICAL FEDERATED LEARNING WITH ATTENTION MECHANISM

In this section, we first define the problem solved in this article, and then propose a metadata and image features co-aware semi-supervised VFL with attention mechanism. To be more precise, by learning the class distribution of existent metadata features, the generator is trained to estimate the missing metadata features in an adversarial manner first. Afterward, in order to train the representation model to extract the representations of features, we map the original features into a latent space based on the label class by minimizing the triple loss function we designed. After that, we estimate pseudo labels for the missing part of party A based on semi-supervised clustering, and then CGAN is used to estimate the missing image features. Based on these foundations, we train the federated model with both metadata and image representations by instructing the model to concentrate on a particular portion of the representation channel with metadata information, which is inspired by the attention mechanism. The detailed framework is depicted in Fig. 3.

A. Problem Definition

Consistent with the typical VFL proposed by Yang et al. [6], we consider a two-party VFL in which only one party owns the label. The primary objective of this scenario is to address the issue that the label-holding party cannot train an accurate model only utilizing local features, so it must instead rely on the other party's features for joint training to improve the accuracy.

In particular, as illustrated in Fig. 2, there are two parties, A and B, with party A having the label and party B lacking it. Party A possesses an image dataset $\mathcal{D}^A = \{(x_i^A, y_i) \,|\, 1 \leq i \leq N^A\}$, where x_i^A and y_i represent the image feature and label of the i^{th} sample, respectively, and N^A is the sample number of $\mathcal{D}^A$. Meanwhile, party B holds a tabular dataset $\mathcal{D}^B = \{x_i^B \,|\, 1 \leq i \leq N^B\}$, where x_i^B denotes the metadata feature of the ith sample and N^B is the sample number of $\mathcal{D}^B$. Raw data cannot be transmitted directly between party A and party B due to the limitations of privacy protection. The aforementioned datasets are further separated based on alignment since there are only partial samples aligned amongst different parties in practice. We assume that the alignment between party A and party B has been implemented according to encrypted entity matching techniques in a privacy-preserving setting described in [43]. The aligned sample set is defined as $\mathcal{D}_{al} = \{(x_i^{A,al}, x_i^{B,al}, y_i^{al}) \,|\, 1 \leq i \leq N^{al}\}$, the part owned by party A is $\mathcal{D}_{al}^A = \{X_{al}^A, Y_{al}\}$ composed of features $X_{al}^A =$

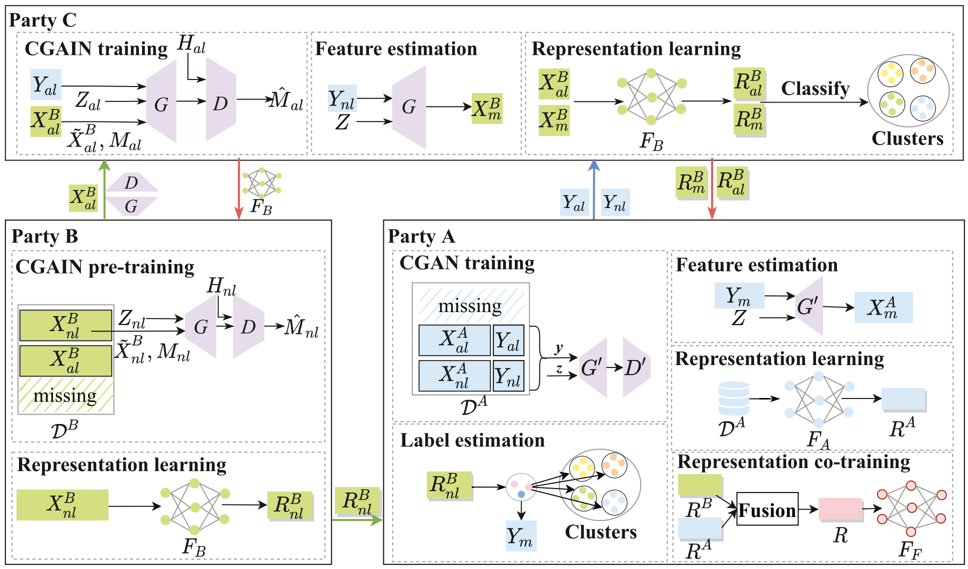


Fig. 3. Framework of MIFC-SVFL.

$\{x_i^{A,al} | 1 \leq i \leq N^{al}\}$ and labels $Y_{al} = \{y_i^{al} | 1 \leq i \leq N^{al}\}$, the part held by party B has only features $\mathcal{D}_{al}^B = X_{al}^B = \{x_i^{B,al} | 1 \leq i \leq N^{al}\}$, and N^{al} is the number of aligned samples. The remaining samples in each party, excluding the aligned samples, are referred to as the non-aligned samples. In party A, these samples are specified as $\mathcal{D}_{nl}^A = \{X_{nl}^A, Y_{nl}\}$, where $X_{nl}^A = \{x_i^{A,nl} | 1 \leq i \leq N_A^{nl}\}$ and $Y_{nl} = \{y_i^{nl} | 1 \leq i \leq N_A^{nl}\}$ are the respective values for features and labels, and as $\mathcal{D}_{nl}^B = X_{nl}^B = \{x_i^{B,nl} | 1 \leq i \leq N_B^{nl}\}$ in party B, N_A^{nl} and N_B^{nl} are the number of non-aligned samples of party A and B, respectively. In addition, the features and labels that do not correspond to the non-aligned samples of the other party are referred to as missing samples.

In the previous VFL schemes, only aligned samples, which are few in number, are used for training, while a large number of non-aligned samples are not used, resulting in data waste and insufficient model training. In addition, the majority of studies solely concentrate on tabular data or image data, and there are few studies that combine these two modalities of data. Therefore, we propose a MIFC-SVFL scheme with attention mechanism, which fully utilizes both aligned and non-aligned samples to further improve the performance of models.

B. Missing Metadata Features Estimation

Some users are reluctant to disclose relevant personal information due to the risk of privacy leakage. In addition, in the case of data aligned between two parties, party A possesses some samples that party B does not. Therefore, there are some missing data in the tabular dataset $\mathcal{D}^B$ of party B. Given the aforementioned issues and motivated by CGAIN proposed by Awan et al. in [44], we estimate the missing features according to the features that are already available.

To maximize the utilization of samples in party B, we pre-train the generator and discriminator in CGAIN with the non-aligned dataset $\mathcal{D}_{nl}^B$. Concretely, the feature set X_{nl}^B is first transformed into a vector $\mathbf{X}_{nl}^B = (x_1^{B,nl}, x_2^{B,nl}, \ldots, x_{N_B^{nl}}^{B,nl})$ and given a random mask vector $\mathbf{M}_{nl} = (M_1, M_2, \ldots, M_{N_B^{nl}})$, where M_i taking values in $\{0, 1\}$, the mask vector is employed to construct a vector containing missing values based on $\mathbf{X}_{nl}^B$ to train CGAIN. Therefore, we define a new vector $\tilde{\mathbf{X}}_{nl}^B = (\tilde{x}_1^{B,nl}, \tilde{x}_2^{B,nl}, \ldots, \tilde{x}_{N_B^{nl}}^{B,nl})$ with (1).

$$\tilde{x}_i^{B,nl} = \begin{cases} x_i^{B,nl}, & M_i = 1, \\ *, & M_i = 0, \end{cases} \tag{1}$$

where symbol * indicates the missing value.

We feed the generator G with the inputs $\mathbf{M}_{nl}$, $\tilde{\mathbf{X}}_{nl}^B$ and random noise $\mathbf{Z}_{nl}$, and the output $\bar{\mathbf{X}}_{nl}^B$ is determined by (2).

$$\bar{\mathbf{X}}_{nl}^B = G(\tilde{\mathbf{X}}_{nl}^B, \mathbf{M}_{nl}, (1 - \mathbf{M}_{nl}) \odot \mathbf{Z}_{nl}), \tag{2}$$

where symbol $\odot$ is element-wise multiplication. What should be emphasized is that $\bar{\mathbf{X}}_{nl}^B$, rather than just the missing part, is the estimation result of all components in $\mathbf{X}_{nl}^B$. The final imputed vector $\hat{\mathbf{X}}_{nl}^B$ is thus obtained from (3).

$$\hat{\mathbf{X}}_{nl}^B = \mathbf{M}_{nl} \odot \tilde{\mathbf{X}}_{nl}^B + (1 - \mathbf{M}_{nl}) \odot \bar{\mathbf{X}}_{nl}^B. \tag{3}$$

The discriminator D is then given a hint vector $\mathbf{H}_{nl}$ indicating whether some values are estimated or original to guide D in determining other values. We input the imputed vector $\hat{\mathbf{X}}_{nl}^B$ and hint vector $\mathbf{H}_{nl}$ into discriminator D to obtain:

$$\hat{\mathbf{M}}_{nl} = D(\hat{\mathbf{X}}_{nl}^B, \mathbf{H}_{nl}), \tag{4}$$

where $\hat{\mathbf{M}}_{nl}$ is the estimated or the raw result that the discriminator D identifies for each value in $\hat{\mathbf{X}}_{nl}^B$.

In the adversarial training process, we expect that the data estimated by the generator can trick the discriminator into believing that it is the original data. At the same time, the estimated data $\hat{x}_i^{B,nl}$ could be as close to the true data $x_i^{B,nl}$ as possible.

Thus, we obtain the loss function:

$$\mathcal{L}_G = \sum_{i=1}^{N_B^{nl}} [(1 - M_i) \log(\hat{M}_i) + M_i^{nl} \log(1 - \hat{M}_i) + \alpha M_i L(x_i, \hat{x}_i)], \quad (5)$$

where α denotes the hyperparameter, L is the distance function.

During the training process, the discriminator aims to clearly distinguish whether the data is estimated or raw. The loss function is shown in (6).

$$\mathcal{L}_D = \sum_{i=1}^{N_B^{nl}} [(1 - M_i) \log(1 - \hat{M}_i) + M_i \log(\hat{M}_i)]. \quad (6)$$

Following the aforementioned training, party B uploads the pretrained generator G, discriminator D, and aligned metadata features X_{al}^B to trusted party C. Meanwhile, party A uploads labels Y_{al} and Y_{nl} to party C.

With the aligned metadata features X_{al}^B and corresponding label Y_{al}, we then train CGAIN. Party C constructs a random mask vector $\mathbf{M}_{al} = (M_1, M_2, \ldots, M_{N^{al}})$ for the vector $\mathbf{X}_{al}^B$ matching to set X_{al}^B. And $\tilde{\mathbf{X}}_{nl}^B = (\tilde{x}_1^{B,al}, \tilde{x}_2^{B,al}, \ldots, \tilde{x}_{N^{al}}^{B,al})$ is obtained based on (1).

After that, we supply the generator G with $\mathbf{M}_{al}$, $\tilde{\mathbf{X}}_{al}^B$, Y_{al} and random noise $\mathbf{Z}_{al}$ to achieve the output specified in (7):

$$\bar{\mathbf{X}}_{al}^B = G(\tilde{\mathbf{X}}_{al}^B | Y_{al}, \mathbf{M}_{al}, (1 - \mathbf{M}_{al}) \odot \mathbf{Z}_{al}). \quad (7)$$

Using (3), we get the imputed vector $\hat{\mathbf{X}}_{al}^B$, and then we enter it along with the hint vector $\mathbf{H}_{al}$ into the discriminator D to obtain the output $\hat{\mathbf{M}}_{al}$ by (8). Then, the loss functions of (5)-(6) are used to train CGAIN.

$$\hat{\mathbf{M}}_{al} = D((\hat{\mathbf{X}}_{al}^B, \mathbf{H}_{al}) | Y_{al}) . \quad (8)$$

Up to this point, the generator has been trained. So that we employ it to estimate the missing features X_m^B in $\mathcal{D}^B$ based on the label Y_{nl} by (9).

$$X_m^B = G(Y_{nl}, \mathbf{Z}). \quad (9)$$

where $\mathbf{Z}$ is the random noise.

C. Metadata Representation Learning

The extraction of data representation has frequently employed neural networks. In order to extract the representation of metadata features $X^B = \{X_{al}^B, X_{nl}^B, X_m^B\}$, a neural network F_B is used to learn the representation R^B from the raw features:

$$R^B = F_B(X^B). \quad (10)$$

When training the neural network F_B, we expect that it will map the raw features into a latent space, where the representations of the same class can be as close as possible, while the representations of different classes can be as far apart as possible, that is, the representations can be classified according to label classes. The corresponding objective function is shown in (11).

$$\min \sum_{i=1}^{N_B^{al}+N_A^{nl}} \sum_{j=1}^{N_B^{al}+N_A^{nl}} d_{ij}, \quad (11)$$

$$d_{ij} = \begin{cases} \left\|R_i^B - R_j^B\right\|_2, & \text{if } y_i = y_j, \\ -\left\|R_i^B - R_j^B\right\|_2, & \text{else,} \end{cases} \quad (12)$$

where $\|\cdot\|_2$ is Euclidean norm.

Motivated by triplet loss [45], the loss that is minimized during training is designed as

$$\mathcal{L}_{R_B} = \sum_{i=1}^{N_B^{al}+N_A^{nl}} \begin{matrix} [\left\|F_B(x_i^B) - F_B(x_{i,p}^B)\right\|_2 \\ -\left\|F_B(x_i^B) - F_B(x_{i,n}^B)\right\|_2 + \beta]_+ \end{matrix}, \quad (13)$$

where $x_{i,p}^B$, $x_{i,n}^B$ are samples of the same class and the different class from sample x_i^B respectively, and β is a margin that is enforced between sample pairs.

In accordance with (11), party C trains the neural network F_B utilizing the aligned features $\mathbf{X}_{al}^B$ and the estimated features X_m^B obtained in the previous section. This results in representations R_{al}^B and R_m^B. Following the training, party C transmits F_B to party B to learn the representations R_{nl}^B of the non-aligned features X_{nl}^B.

D. Missing Image Features Estimationn

When party A and party B are aligned, party B has some features that party A is missing. As a result, we train CGAN to estimate the missing part in party A.

In the beginning, party A trains CGAN with its local labeled dataset $\mathcal{D}^A$. In order to teach a generator to produce images according to given labels, discriminator and generator are training in an adversarial manner. The corresponding objective function is shown in (14).

$$\min_{G'} \max_{D'} V(D', G') = \mathbb{E}_{x\sim\mathcal{D}^A}[\log D'(x|y)] + \mathbb{E}_{z\sim p_z(z)}[\log(1 - D'(G'(z|y)))], \quad (14)$$

where G', D' are the generator and discriminator of CGAN respectively, $p_z(z)$ is the prior on input noise variables z.

For the generator, it is dedicated to generating images of given labels to fool the discriminator into believing it is raw, while the training task of the discriminator is to become more adept at determining whether the input data is raw or synthetic. Therefore, the loss functions of discriminator and generator are shown in (15) and (16), respectively.

$$\mathcal{L}_{D'} = \log(1 - D'(G'(z|y))) - \log(D'(G'(z|y))), \quad (15)$$

$$\mathcal{L}_{G'} = -\log(D'(x|y)). \quad (16)$$

E. Metadata and Image Representations Co-Training

Party A being the party with labels signifies that both features and labels are absent from the missing part. Therefore, party B transmits the non-aligned representations R_{nl}^B to party A to estimate the pseudo labels of this missing part. To be more specific, party A executes classification on the remaining representations R_{al}^B and R_m^B of party B that downloaded from party C. According to the analysis in the previous section, these representations will be grouped into K clusters in accordance with the labels. Consequently, we measured the distances between the estimated

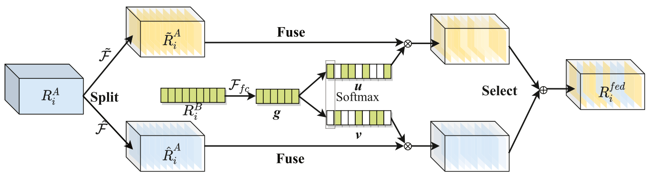


Fig. 4. Framework of fusing metadata and image representations with SKNets.

representation R^B_{nl} and these clusters and chose the cluster with the shortest distance as its corresponding label, i.e., (17).

$$y_i^m = \underset{k\in\{1,2,\cdots K\}}{\arg\min} \left\| R_i^{B,nl} - c_k \right\|_2, 1 \leq i \leq N_B^{nl}, \tag{17}$$

where c_k is the center point of the kth cluster, K is the total number of label classes of X^B. According to (17), we obtain the pseudo labels Y_m of this missing part and put them into the generator that has been trained using (15)-(16) to estimate the missing image features:

$$X_m^A = G'(Y_m, Z). \tag{18}$$

After that, the neural network F_A receives the supplementary dataset $\mathcal{D}^A = \mathcal{D}^A \cup X_m^A \cup Y_m$ for supervised training, and the relevant representations R^A are made according to the (11).

In the existing researches, the fusion of metadata representations and image representations is frequently done directly in a straightforward vector concatenation manner, which is then sent to the full connection layer for training. Nevertheless, the simple concatenation overlooks the hidden influence of metadata on image features. As an instance, metadata can be utilized in skin lesion diagnosis to restrict the predicted probability of other irrelevant skin lesions, when a certain body location of disease is only associated with only a few skin diseases [46]. Inspired by SKNets attention mechanism [32], as depicted in Fig. 4, we propose a method of fusing metadata and image representations with attention mechanism, employing metadata to manage the importance of each representation channel, and allowing the network to concentrate on a particular portion of the representation channel with specific metadata information.

Firstly, any image representation R_i^A is converted to $\tilde{R}_i^A$ and $\hat{R}_i^A$ by two transformations $\tilde{\mathcal{F}}$ and $\hat{\mathcal{F}}$ of different kernel sizes (such as 3 and 5), respectively. Note that we only show a two-branch example here, but one might easily infer instances with more branches. Subsequently, the corresponding metadata representation R_i^B provides guidance for accurate and adaptive selections by reducing dimension through the full connection layer $\mathcal{F}_{fc}(\cdot)$, and obtains the representation:

$$g = \mathcal{F}_{fc}(R_i^B) = \delta(\mathcal{B}(\mathbf{W}R_i^B)), \tag{19}$$

where δ represents the ReLU activation function, $\mathcal{B}$ is the batch normalization, and $\mathbf{W}$ denotes the weight matrix.

Additionally, cross-channel soft attention is used for g to adaptively select information from various spatial scales. Concretely, apply softmax operator on channel-wise digits:

$$u_j = \frac{e^{U_j g}}{e^{U_j g} + e^{V_j g}}, v_j = \frac{e^{V_j g}}{e^{U_j g} + e^{V_j g}}, 1 \leq j \leq J, \tag{20}$$

$$u_j + v_j = 1, \tag{21}$$

where J is the number of channels of image representation, u and v denote the soft attention vector for $\tilde{R}_i^A$ and $\hat{R}_i^A$, respectively, and U, V are weight matrices. Notice that the subscript j represents the jth row of the matrix or the jth element of the vector.

Thus, the attention weighted sum of various kernels generates the final fusion representation R:

$$R_{i,j} = u_j \cdot \tilde{R}_{i,j}^A + v_j \cdot \hat{R}_{i,j}^A, \tag{22}$$

$$R_i = [R_{i,1}, R_{i,2}, \ldots, R_{i,J}]. \tag{23}$$

Eventually, we feed the representations R which are fused by metadata and image representations into the federated model F_F for training. The relevant cross-entropy loss function is:

$$\mathcal{L}_R = -\sum_{i=1}^{N^A} y_i \log(F_F(R_i)). \tag{24}$$

To provide a better grasp of the method proposed in this research, we summarize it as Algorithm 1.

F. Convergence Analysis

In order to verify the effectiveness of the proposed algorithm, we provide the convergence analysis for Algorithm 1 in this subsection. Based on the procedures of Algorithm 1, we can obtain the following convergence result as Theorem 1.

Theorem 1: If representation vectors $F(x_i), F(x_{i,p}), F(x_{i,n})$ are all bounded, then Algorithm 1 possesses the characteristic of convergence.

Proof: The algorithm mainly consists of four procedures: missing metadata features estimation, metadata representation learning, missing image features estimation and representation learning, and metadata and image representations co-training. Our convergence proof is mainly developed from the above four procedures.

Procedure 1: Missing metadata features are estimated by the GAIN in reference [44], which has been verified the convergence and effectiveness. Therefore, the high quality of synthetic data generated based on GAIN can ensure the convergence of the model.

Procedure 2: Based on the high-quality synthetic metadata generated by procedure 1, procedure 2 uses the loss function in (13) and stochastic gradient descent (SGD) to train model F_B to extract the representations of metadata features. According to the existing knowledge, when using SGD to train the model, the following three conditions need to be met to ensure the model convergence: 1) The gradient of the loss function meets Lipschitz continuity; 2) The loss function has a lower bound; and 3) The suitable learning rate. Since the choice of learning rate in the third condition primarily influences the convergence speed, the effect for model convergence is limited in essence, hence we ignore the discussion of learning rate in the following proof. Furthermore, we omit the notation B in (13) in the following verification.

1) *Assumption 1:* We assume that vectors $F(x_i)$, $F(x_{i,p})$, $F(x_{i,n})$ are all bounded.

Based on the above assumption, the gradient for (13) is obtained as follows:

$$\begin{aligned}&\nabla \mathcal{L}_R(F(x_i), F(x_{i,p}), F(x_{i,n}))\\&= \left(\frac{\partial \mathcal{L}_R}{\partial F(x_i)}, \frac{\partial \mathcal{L}_R}{\partial F(x_{i,p})}, \frac{\partial \mathcal{L}_R}{\partial F(x_{i,n})}\right)\\&= (2(F(x_{i,n}) - F(x_{i,p})), 2(F(x_{i,p}) - F(x_i)),\\&\quad 2(F(x_i) - F(x_{i,n}))). \end{aligned} \tag{25}$$

Therefore, $\nabla \mathcal{L}_R$ is bounded as long as Assumption 1 is true, thus $\nabla \mathcal{L}_R$ meets Lipschitz continuity because the bound function is a special Lipschitz continuity function.

2) We can learn the following result from the (13), i.e., if $\|F(x_i) - F(x_{i,p})\|_2 > \|F(x_i) - F(x_{i,n})\|_2 - \beta$, then $\nabla \mathcal{L}_R > 0$, otherwise $\nabla \mathcal{L}_R = 0$. Therefore, $\mathcal{L}_R \geq 0$, which means $\mathcal{L}_R$ has a lower bound.

In summary, if vectors $F(x_i)$, $F(x_{i,p})$, $F(x_{i,n})$ are all bounded, we can ensure the convergence of procedure 2.

Procedure 3: The estimation of missing image features is mainly based on GAN, and its convergence and effectiveness have also been verified in reference [47]. Therefore, the quality of synthetic data generated based on GAN can ensure the convergence of the model. Next, the model F_A is trained using the loss function in (13) and SGD to extract the representation of the features. The convergence of representation learning can be ensured, and its analysis process is similar to procedure 2.

Procedure 4: After representations of the entire sample space are obtained, we use both metadata and image representations to train federated model. In this procedure, we mainly use the cross-entropy loss and SGD to train model. These traditional loss functions, such as cross-entropy loss and mean squared error, have been proved that can ensure the convergence of the model. Although we introduce the attention mechanism, which uses metadata representations to emphasize important parts of image representations to improve model performance. Since the attention mechanism is embedded in the model structure to improve the model convergence effect, so it will not affect whether the model can converge. Therefore, we can ensure the convergence of the entire procedure 4.

According to the above analysis, we find that as long as vectors $F(x_i)$, $F(x_{i,p})$, $F(x_{i,n})$ are bounded, these four procedures can converge, so that Algorithm 1 is convergent. ∎

Algorithm 1: MIFC-SVFL with attention mechanism.

Input: Datasets $\mathcal{D}^A$ and $\mathcal{D}^B$; Neural networks F_A, F_B, and F_F; Generators G and G'; Discriminators D and D'; Epoch number T.
Output: Federated model F_F.

BEGIN
for $t = 1, 2, \cdots, T$ **do**
  procedure MISSING METADATA FEATURES ESTIMATION
    Pre-train G and D with X_{nl}^B by minimizing (5)–(6) in party B;
    Party B uploads the trained G, D and aligned features X_{al}^B to party C;
    Party A uploads labels Y_{al} and Y_{nl} to party C;
    Train CGAIN with the aligned data according to (7)–(8) and minimize (5)–(6);
    Estimate the missing features X_m^B by (9);
  procedure METADATA REPRESENTATION LEARNING
    Train F_B to learn the representations R^B based on classes with (13);
    Party C transmits F_B to party B to learn R_{nl}^B by (10);
  procedure MISSING IMAGE FEATURES ESTIMATION AND REPRESENTATION LEARNING
    Party A trains CGAN with its local dataset $\mathcal{D}^A$ by (14)–(16);
    Estimate the pseudo labels Y_m of this missing part according to (17);
    Estimate the missing image features through the trained G';
    Train F_A to learn the representations R^A with (13);
  procedure METADATA AND IMAGE REPRESENTATIONS CO-TRAINING
    Convert R_i^A into $\tilde{R}_i^A$ and $\hat{R}_i^A$ by two transformations $\tilde{\mathcal{F}}$ and $\hat{\mathcal{F}}$;
    Reduce dimension of R_i^B through (19);
    Select information from various spatial scales by (20)–(21);
    Generate the final fusion representations R with (22)–(23);
    Train the federated model F_F through (24).
END

V. EXPERIMENTAL RESULTS AND EVALUATION

In this section, relevant simulation experiments are carried out to evaluate the performance of the proposed MIFC-SVFL scheme, and the effectiveness and superiority of our scheme are demonstrated by comparison with other related works in the VFL scenario.

A. Experimental Setup

Datasets setting: We primarily employ two different datasets to conduct experiments. 1) **PAD-UFES-20 dataset** [48]: This dataset contains 2,298 samples of six different classes of skin diseases, each containing a clinical image taken by smartphones as well as relevant patient information that includes 21 tabular features, such as age, gender, cardinal symptoms, etc. In consideration of the integrity of the tabular data and their relevance to the diagnosis of skin lesions, we only utilize the remaining 17 tabular features after removing "background_father", "background_mother", "diameter_1", and "diameter_2". 2) **ISIC 2019** [49], [50], [51]: This dataset includes 25,331 samples from eight various classes of skin lesions, each of which has a dermoscopic image and three metadata features indicating the age, sex and general anatomic site of patients. To simulate the VFL scenario, we divided the image features and labels from the dataset into party A, the remaining tabular features into party B.

In addition, each dataset is randomly split into a training set and a testing set with a ratio of 8:2. When simulating the aligned and non-aligned sample sets in subsequent experiments, we make an effort to keep the ratio of missing image features and tabular features the same. For instance, if the training dataset contains 2000 samples overall and 1000 of them are aligned, then there are 500 missing samples of both the image and tabular features.

Model setting: For the extraction of tabular features, we employ a simple neural network composed of three linear layers and corresponding ReLU activation functions. Meanwhile, we apply the SENet-154 neural network structure, which has been pre-trained on ImageNet, for image features and fused features based on the attention mechanism.

Hyperparameter and configuration: We set the hyperparameter $\alpha = 0.1$ in (5) and $\beta = 0.3$ in (13). All the experiments are implemented by Pytorch and trained on Nvidia GeForce RTX 3090 GPU with 24 GB GDDR6X memory.

Baseline models: In order to further verify the effectiveness and superiority of the proposed scheme, we compare MIFC-SVFL with other related works in the VFL scenario. 1) **Vanilla-VFL**: Only the aligned samples between party A and party B are taken, while the non-aligned samples are ignored. 2) **FedCVT** [21]: To expand the aligned dataset, the representations of missing features are estimated based on the weighted sum of existing representations, and the pseudo-labels of unlabeled samples are predicted with cross-view. 3) **FedMC** [22]: Based on overlapping samples, a common feature space is built, and the distance between each pair of non-overlapping samples in this space is calculated to match and enrich the dataset. Since the original literatures of the above methods do not provide the original code, the experiment of comparison experiments is carried out according to the description in the original papers.

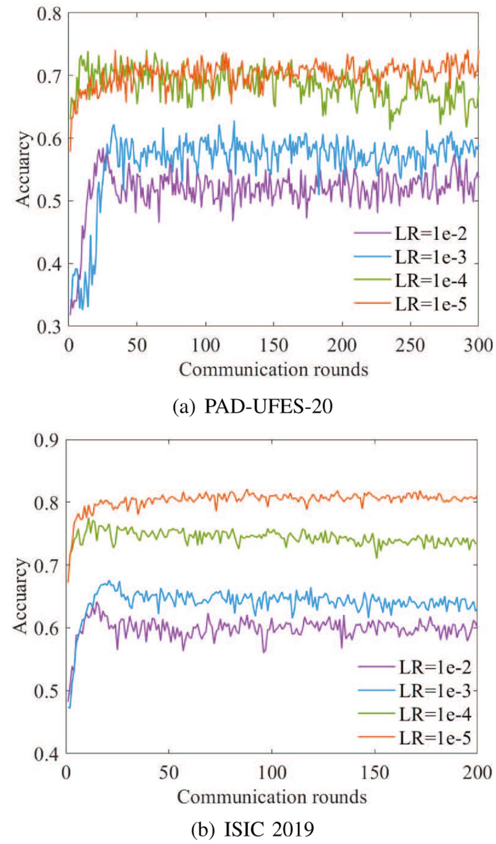


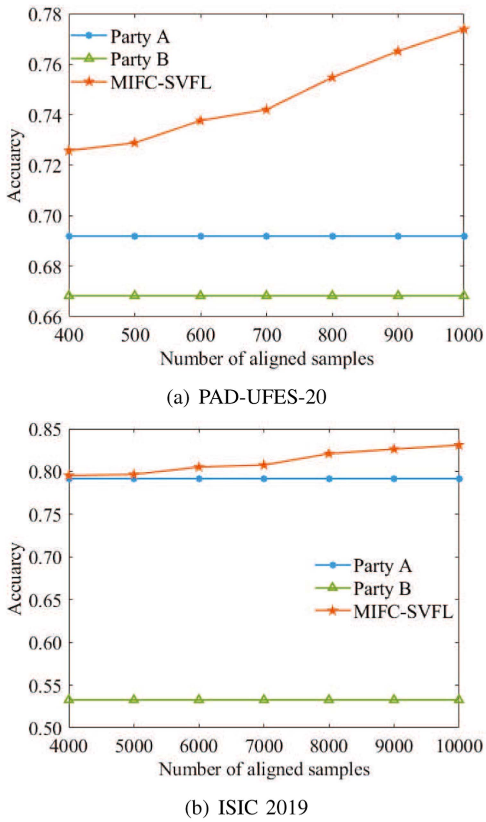
Fig. 5. Accuracy evaluation with the increase of communication rounds. (a) PAD-UFES-20. (b) ISIC 2019.

B. Results and Comparisons

The verification of the convergence initiates the experiment. Fig. 5(a) and (b) respectively depict the variation curves of accuracy with the increase of communication rounds under different learning rates (LR) on PAD-UFES-20 and ISIC 2019. It can be seen that as the communication rounds increase, accuracy does as well until it eventually tends to converge, which validates the effectiveness of the proposed MIFC-SVFL. In addition, our proposed scheme exhibits certain advantages in terms of convergence speed. Specifically, it can attain convergence within 50 communication rounds on PAD-UFES-20 and within 30 communication rounds on ISIC 2019. Moreover, when LR is lower, the amplitude of each parameter update is smaller, resulting in a more stable training process and higher accuracy. Therefore, we set LR = 1e-5 in subsequent experiments.

The accuracy of local training and joint training of party A and party B with two different datasets are depicted in Fig. 6(a) and (b), respectively. It is apparent that the local training of party A performs with greater accuracy than that of party B, indicating that image features are more discriminative than tabular features. Furthermore, it is evident that the accuracy of the joint training is significantly higher than that of the two features trained separately, and that a noticeable performance improvement can be obtained with the increase of the number of aligned samples. The reason for this is that the VFL model can fully utilize the features of both image and tabular modalities for joint training without violating privacy, boosting the performance of the model. Additionally, the more aligned samples there are between the parties, the more references there are for estimating the missing sample representations, and consequently, the more accurate the prediction outcomes will be. Meanwhile, since there are only three metadata features in the ISIC 2019 dataset, the latent impact of metadata on images is relatively small compared to the PAD-UFES-20 dataset, however Fig. 6(b) still demonstrates considerable performance gains.

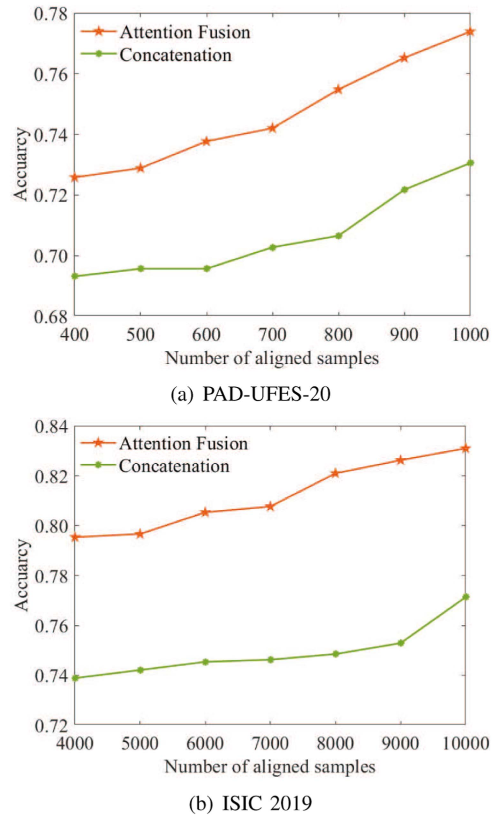
Fig. 7(a) and (b) visualize the accuracy comparison curves on PAD-UFES-20 and ISIC 2019 using simple concatenation and attention fusion for joint training in VFL. As can be seen, the proposed joint training strategy based on attention fusion is unquestionably superior than the already well-liked simple concatenation (about 3.2%–4.8% on PAD-UFES-20, about 5.5%-7.3% on ISIC 2019). This is due to the fact that for a sample, its metadata features have certain potential effects on image features. As a result, the attention fusion strategy can leverage metadata features to highlight critical parts of image features that are beneficial for classification, increasing the performance of model. Simple concatenation, however, assumes that the two are independent and ignores the potential relationship between them, making it impossible to produce the desired result.



(a) PAD-UFES-20

(b) ISIC 2019

Fig. 6. Accuracy of local training and joint training of party A and party B. (a) PAD-UFES-20. (b) ISIC 2019.



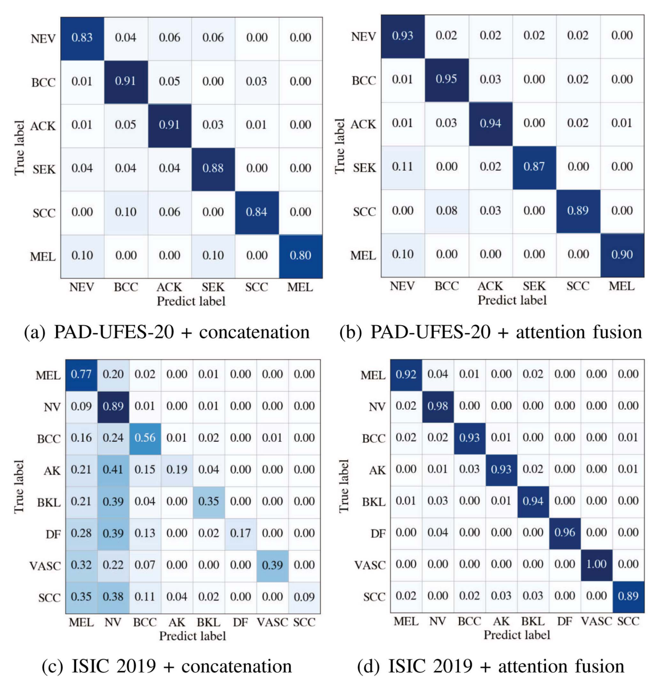
(a) PAD-UFES-20

(b) ISIC 2019

Fig. 7. Accuracy comparison of concatenation and attention fusion for joint training in VFL. (a) PAD-UFES-20. (b) ISIC 2019.

Detailed classification results of each category are shown in the confusion matrix in Fig. 8. As seen in the figure, attention fusion on both datasets achieves superior accuracy over simple concatenation. Only the SEK category of the PAD-UFES-20 dataset shows a 1% lower accuracy with attention fusion compared to concatenation, whereas the other categories all achieve a 3% to 10% superior accuracy. Moreover, the performance of attention fusion on ISIC 2019 significantly exceedes that of concatenation. The poor performance shown in Fig. 8(c) is mainly due to the extreme class imbalance in the ISIC 2019 dataset, which indirectly indicates that our method has certain effectiveness in addressing class imbalance. These experiments serve as additional evidence of the superiority of our proposed collaborative training based on attention mechanism.

In order to further demonstrate the superiority of our scheme, Fig. 9(a) and (b) exhibit the curves of the accuracy of various VFL methods on two datasets as the number of aligned samples increases. As Vanilla-VFL only employs aligned samples, which are a small quantity for FL training, it should come as no surprise that accuracy is extremely low and even inferior to party B's local training performance in Fig. 9(a). The accuracy of the other schemes, which utilize both aligned and non-aligned samples, is higher than the local training results of party A and party B, demonstrating that maximizing the usage of the samples from



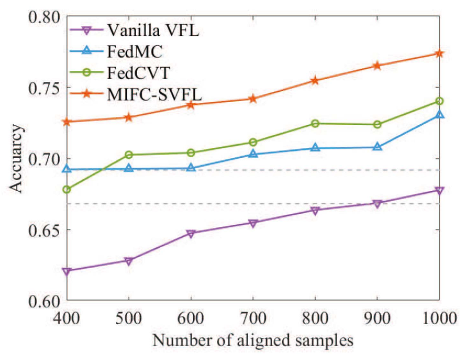
(a) PAD-UFES-20 + concatenation (b) PAD-UFES-20 + attention fusion

(c) ISIC 2019 + concatenation (d) ISIC 2019 + attention fusion

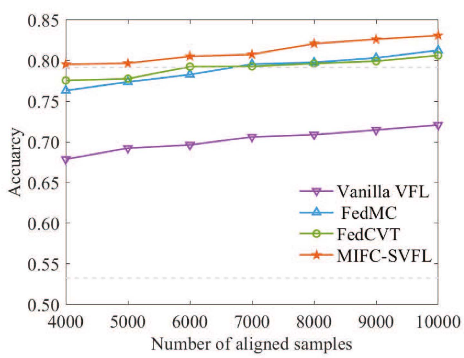
Fig. 8. Confusion matrices of concatenation and attention fusion method on different datasets. (a) PAD-UFES-20 + concatenation. (b) PAD-UFES-20 + attention fusion. (c) ISIC 2019 + concatenation. (d) ISIC 2019 + attention fusion.

TABLE II
THE RESULTS OF KEY COMPONENTS IN MIFC-SVFL

Dataset	Component	400	500	600	700	800	900	1000
PAD-UFES-20	Baseline	0.6210	0.6284	0.6476	0.6550	0.6638	0.6687	0.6779
	Baseline + MME	0.6436	0.6480	0.6592	0.6631	0.6756	0.6814	0.6945
	Baseline + MIE	0.6672	0.6711	0.6779	0.6842	0.6932	0.7141	0.7197
	Baseline + ME	0.6931	0.6957	0.6966	0.7027	0.7065	0.7217	0.7305
	Baseline + AF	0.6690	0.6722	0.6889	0.6882	0.6987	0.7194	0.7355
	Baseline + ME+AF	0.7258	0.7289	0.7377	0.7420	0.7547	0.7652	0.7739
Dataset	**Component**	**4000**	**5000**	**6000**	**7000**	**8000**	**9000**	**10000**
ISIC 2019	Baseline	0.6788	0.6923	0.6965	0.7061	0.7090	0.7146	0.7209
	Baseline + MME	0.6952	0.7102	0.7183	0.7265	0.7298	0.7351	0.7417
	Baseline + MIE	0.7103	0.7294	0.7305	0.7386	0.7401	0.7479	0.7635
	Baseline + ME	0.7389	0.7421	0.7454	0.7463	0.7485	0.7529	0.7714
	Baseline + AF	0.7092	0.7265	0.7372	0.7423	0.7570	0.7640	0.7745
	Baseline + ME+AF	0.7954	0.7966	0.8053	0.8076	0.8210	0.8262	0.8310



(a) PAD-UFES-20



(b) ISIC 2019

Fig. 9. Accuracy comparison of different methods. (a) PAD-UFES-20. (b) ISIC2019.

the entire sample space is more beneficial for FL significant performance. On this basis, we can find that our proposed scheme has advantages over FedCVT and FedMC (about 3%–6% on PAD-UFES-20, about 1.3%–2.7% on ISIC 2019). This is largely due to the estimation of the missing representations and the bimodality features fusion method based on attention mechanism, which allows VFL to utilize more samples and explore the deep relations between the two modalities.

Ablation experiments are carried out on two datasets, PAD-UFES-20 and ISIC 2019, with various numbers of aligned samples to show the effectiveness of each component in MIFC-SVFL. Using Vanilla-VFL as the baseline, the performance improvement of MIFC-SVFL is primarily made up of two components: missing representation estimation (ME) and attention fusion (AF). In order to show the effectiveness of each proposed module, ME can be further specified as missing metadata representation estimation (MME) and missing image representation estimation (MIE). It can be seen from Table II that the application of these two components can improve the model performance. When the number of aligned samples is small, the ME can fundamentally broaden the training sample space. Therefore, using ME alone performs better than using AF alone. Furthermore, both MME and MIE can improve performance on their own, with MIE performing better than MME as a result of image features having a higher level of discrimination than metadata features. However, as the number of aligned samples increases, using only AF can surpass the performance of using only ME. This further demonstrates the effectiveness of the attention mechanism in exploring the implicit relationships between features. There is no denying that combining this two can boost performance even further.

VI. CONCLUSION

In this work, we construct a metadata and image features co-aware VFL model for smart healthcare, from which users can obtain auxiliary diagnostic results. Furthermore, we propose a MIFC-SVFL scheme with attention mechanism, which aims to jointly train the VFL models by utilizing both metadata and image data from the entire sample space. Specifically, a generate-adversarial method is designed to train the generator to estimate the missing features, so as to expand the training set. Moreover, using semi-supervised clustering to forecast pseudo labels for the missing part of the active party, we further boost the performance of the VFL model. Meanwhile, we establish an entirely novel VFL training mode that makes metadata features highlight crucial areas in the image features to improve the training effect. Extensive experiments on two different datasets confirm that our proposed scheme achieves better accuracy compared with other related works in the VFL scenario. In the future, we plan to extend this work into multi-modality VFL scenarios.

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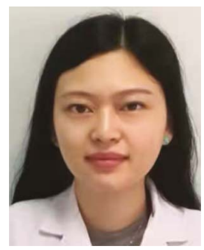

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